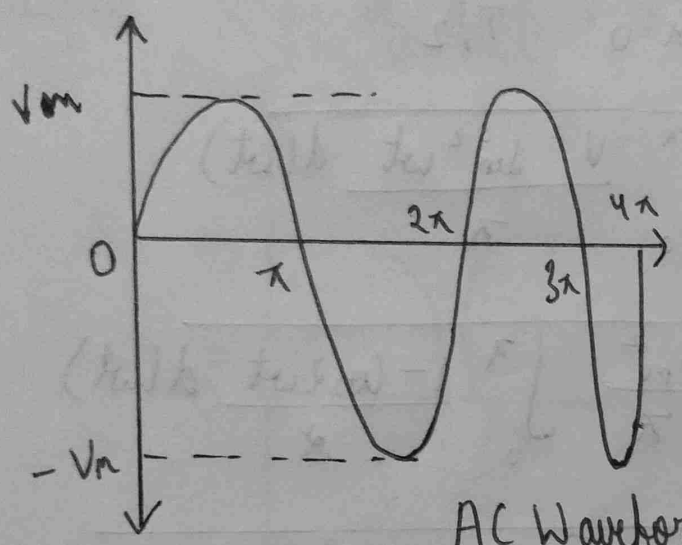
Unit - 02

Important Question

Q) Derive average & RMS Value, form factor, peak factor of sine Wave



AC Waveform

$T = 2\pi$

[Waveform is repeating from Here]

$V = V_m \sin \omega t$

$$\boxed{V_{av} = \int_{\omega t=0}^{T/2} \frac{V\, d\omega(t)}{T/2}}$$

$$V_{av} = \int_0^{\pi} \frac{V_m \sin \omega t \; d(\omega t)}{\pi}$$

$$V_{av} = \frac{V_m}{\pi} \int_0^{\pi} \sin \omega t \; d(\omega t)$$

$$= \frac{V_m}{\pi} \left[-\cos \omega t \right]_{\omega t=0}^{\omega t=\pi}$$

$$= \frac{V_m}{\pi} \left[-\cos \pi + \cos 0 \right]$$

$$V_{av} = \frac{V_m}{\pi}\left[1+1\right] = \frac{2V_m}{\pi}$$

$$\boxed{V_{av} = \frac{2V_m}{\pi}}$$

$$V_{RMS} = \sqrt{\int_{\omega t=0}^{T/2} \frac{V^2\, d(\omega t)}{T/2}}$$

$$V_{RMS} = \sqrt{\int_0^{\pi} \frac{V^2 \sin^2 \omega t}{\pi}\, d(\omega t)}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \frac{1-\cos 2\omega t}{2}\, d(\omega t)}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{\pi}\left[\frac{\pi}{2} - \frac{\sin 2\pi}{4} + \frac{\sin 0}{4}\right]}$$

$$\boxed{V_{RMS} = \frac{V_m}{\sqrt{2}}}$$

Form factor :- $K_f = \dfrac{\text{RMS Value}}{\text{Average Value}}$

$$= \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} = \frac{\pi}{2\sqrt{2}}$$

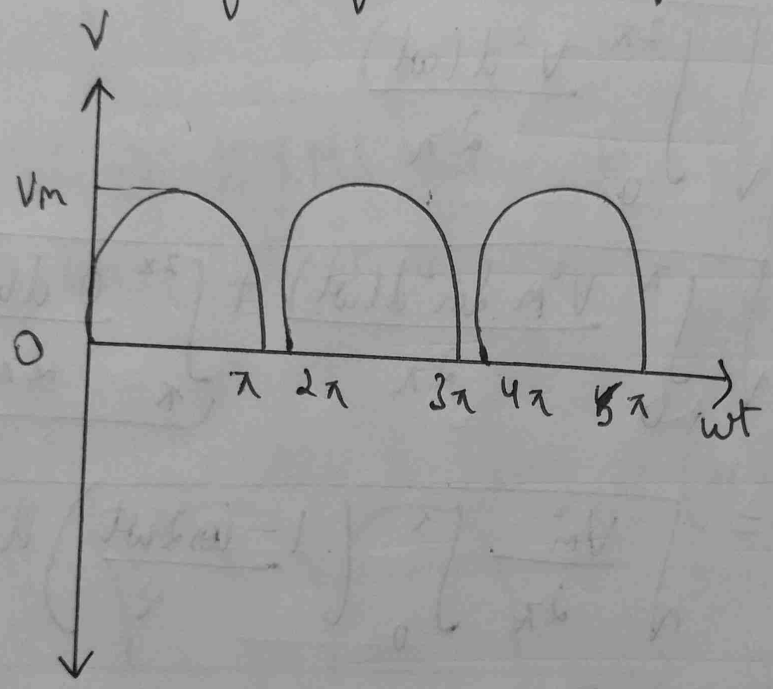
$$\boxed{K_f = 1.11}$$

Peak factor

$$K_p = \frac{V_{max}}{V_{RMS}} = \frac{\cancel{V_m}}{\frac{\cancel{V_m}}{\sqrt{2}}} = \sqrt{2}$$

$$\boxed{K_p = 1.41}$$

Q) Derive average & RMS Value, form factor, peak factor of output wave of half wave Rectifier.



$T = 2\pi$

DC Waveform

$V = V_m \sin\omega t$

$0 < \omega t < \pi$

$V = 0$

$\pi < \omega t < 2\pi$

Average Value:

$$V_{av} = \int_{\omega t=0}^{\pi} \frac{V\,d(\omega t)}{T}$$

$$V_{av} = \int_{\omega t=0}^{2\pi} \frac{V\,d(\omega t)}{2\pi}$$

$$V_{av} = \int_{\omega t=0}^{\pi} \frac{V_m \sin\omega t\, d(\omega t)}{2\pi} + \underbrace{\int_{\pi}^{2\pi} \frac{0\, d(\omega t)}{2\pi}}_{0}$$

$$V_{av} = \frac{V_m}{2\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$V_{av} = \frac{V_m}{2\pi}\left[-\cos\pi + \cos 0\right]$$

$$V_{av} = \frac{2V_m}{2\pi} = \frac{V_m}{\pi}$$

RMS Value $\quad V_{RMS} = \sqrt{\int_0^T \frac{V^2 d(\omega t)}{T}}$

$$V_{RMS} = \sqrt{\int_0^{2\pi} \frac{V^2 d(\omega t)}{2\pi}}$$

$$V_{RMS} = \sqrt{\int_0^{\pi} \frac{V_m^2 \sin^2 d(\omega t)}{2\pi} + \underbrace{\int_{\pi}^{2\pi} \frac{0\, d(\omega t)}{2\pi}}_{\to 0}}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi}\left(\frac{1-\cos 2\omega t}{2}\right) d(\omega t)}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

$$V_{RMS} = \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\pi}{2}\right]}$$

$$\boxed{V_{RMS} = \frac{V_m}{2}}$$

Form factor :- $\quad \frac{\text{RMS Value}}{\text{Average Value}} = K_f$

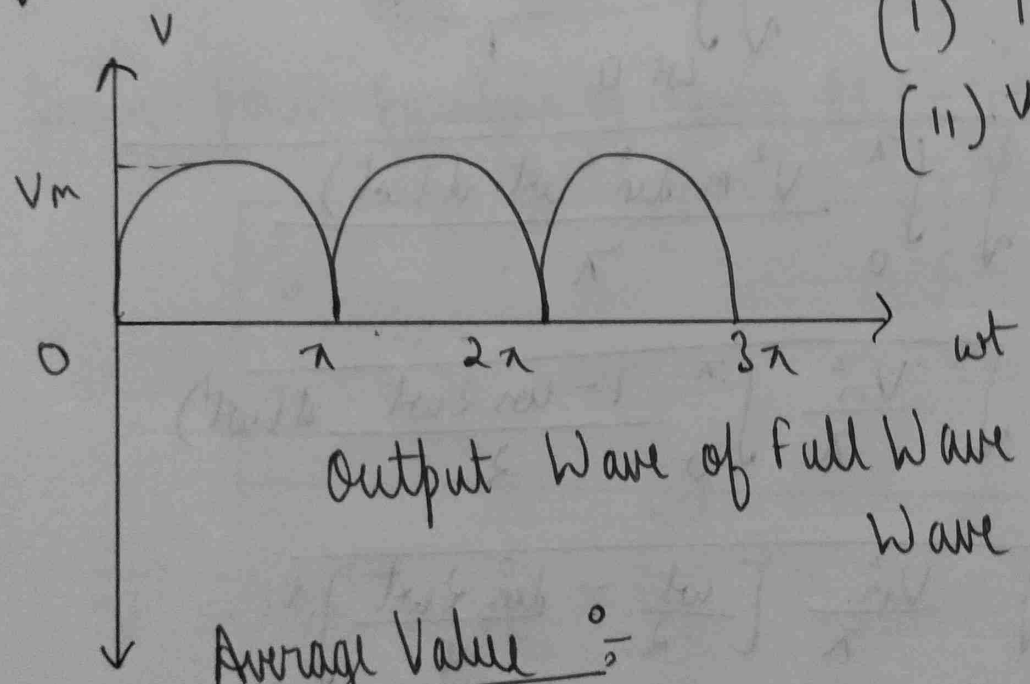
$$K_f = \frac{\frac{V_m}{2}}{\frac{V_m}{\pi}} = \frac{\pi}{2} = \boxed{1.57 = K_f}$$

Peak factor

$$K_p = \frac{V_{max}}{V_{rms}} = \frac{V_m}{\frac{V_m}{2}}$$

$$\boxed{K_p = 2}$$

Q) Derive average & RMS value, form factor, peak factor of full wave rectifier.



(I) $T = \pi$

(II) $V = V_m \sin \omega t$

$0 < \omega t < \pi$

Output Wave of Full Wave Rectifier DC Wave

<u>Average Value</u> :-

$$V_{av} = \int_0^T \frac{V \, d(\omega t)}{T}$$

$$V_{av} = \int_0^{\pi} \frac{V_m \sin \omega t \; d(\omega t)}{\pi}$$

$$V_{av} = \frac{V_m}{\pi}\left[\cos \omega t\right]_0^{\pi}$$

$$V_{av} = \frac{V_m}{\pi}\left[\cos \pi - \cos 0\right]$$

$$V_{av} = \frac{V_m}{\pi}\left[1+1\right]$$

$$\boxed{V_{av} = \frac{2V_m}{\pi}}$$

<u>RMS Value</u> :

$$V_{RMS} = \sqrt{\int_{\omega t=0}^{T} \frac{V^2 d(\omega t)}{T}}$$

$$V_{RMS} = \sqrt{\int_0^{\pi} \frac{V^2 m \sin^2 \omega t \, d(\omega t)}{\pi}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \frac{1-\cos 2\omega t}{2} d(\omega t)}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left[\frac{\pi}{2} - \frac{\sin 2\pi}{4} - \frac{0}{2} + \frac{\sin 0}{2}\right]}$$

$$\boxed{V_{RMS} = \frac{V_m}{\sqrt{2}}}$$

Form factor :-

$$K_f = \frac{\text{RMS Value}}{\text{AV Value}} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}}$$

$$K_f = \frac{\pi}{2\sqrt{2}}$$

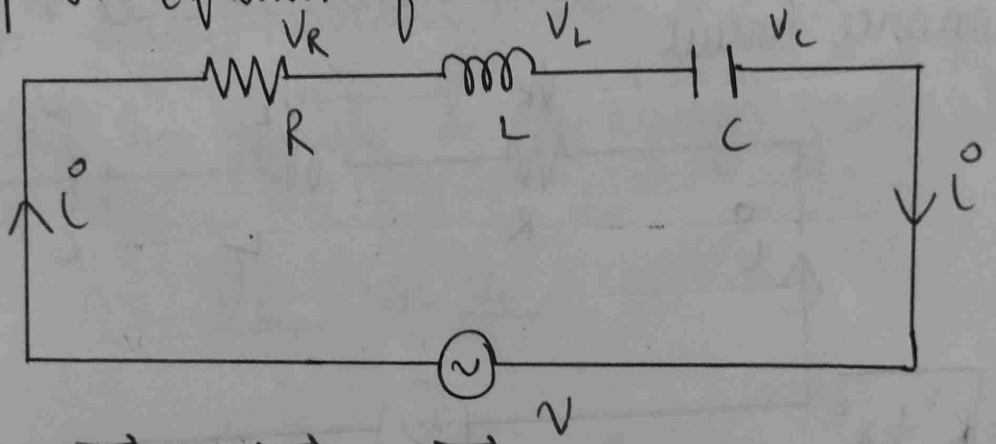
$$\boxed{K_f = 1.11}$$

Peak factor :-

$$K_p = \frac{V_m}{V_{RMS}} = \frac{\cancel{V_m}}{\frac{\cancel{V_m}}{\sqrt{2}}} = \sqrt{2}$$

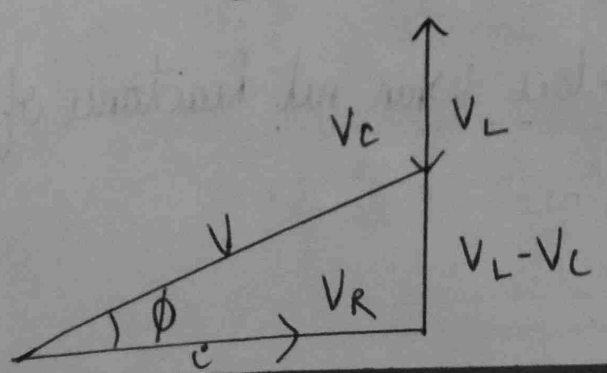
$$\boxed{K_p = 1.41}$$

Q) Derive power Equation of Series RLC Circuit.

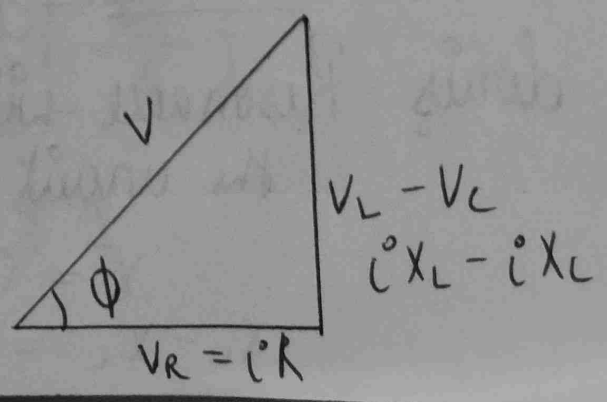


$$\vec{V} = \vec{V_R} + \vec{V_L} + \vec{V_C}$$

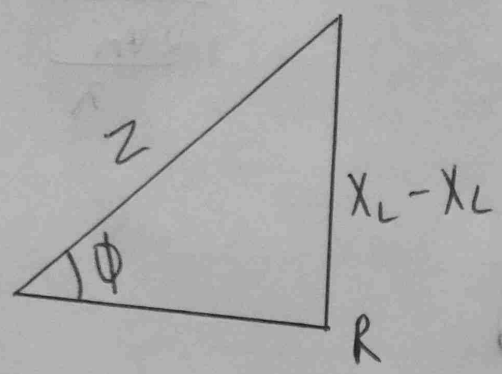
Phasor Diagram



Voltage Diagram



Impedance Triangle



Average Power.

$$|V| = \sqrt{(V_R)^2 + (V_L - V_C)}$$

$$|V| = \sqrt{R^2 + (X_L - X_C)^2}$$

$$|V| = i\,|Z|$$

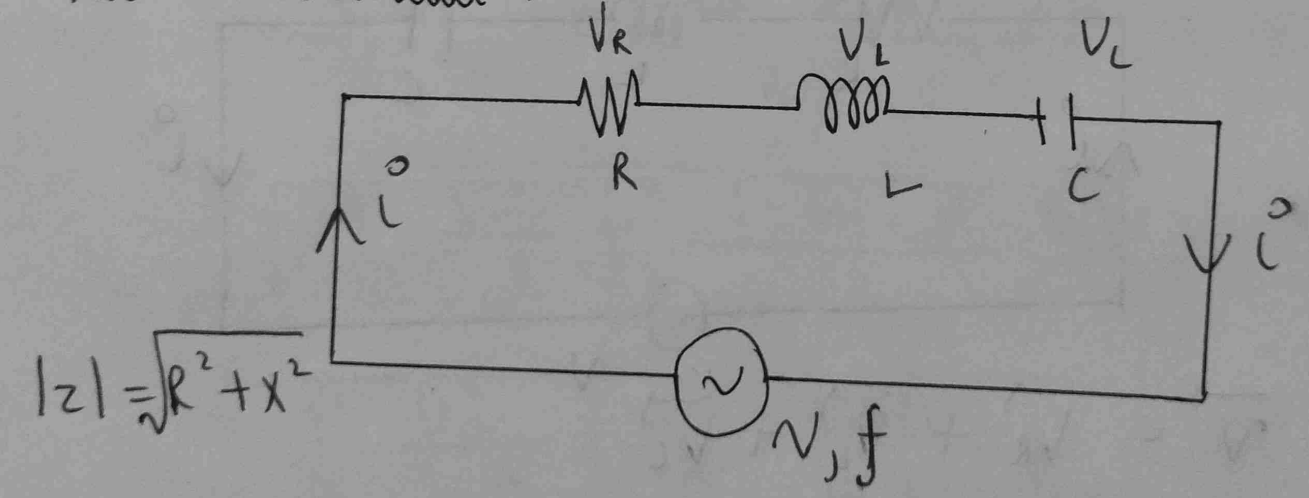
$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\boxed{P_{av} = V_r I_r \cos\phi}$$

Power factor

$$\cos\phi = \frac{R}{Z}$$

Q) Derive Resonance frequency & quality factor of series Resonance Circuit.



Resonance frequency

Series Resonance will take place when net Reactance of the circuit is zero.

$$X = 0$$

$$X_L - X_C = 0$$

$$W_r L - \frac{1}{W_r C} = 0$$

$$W_r L = \frac{1}{W_r C}$$

$$W_r^2 = \frac{1}{LC}$$

$$\boxed{W_r = \frac{1}{\sqrt{LC}}}$$ Angular Resonance frequency

$$\boxed{f_r = \frac{W_r}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}}$$ Resonance frequency

<u>Quality factor</u>:- Quality factor is the voltage magnification in series Resonance Circuit.

It is a Unit less quantity & it always greater than & Equal to.

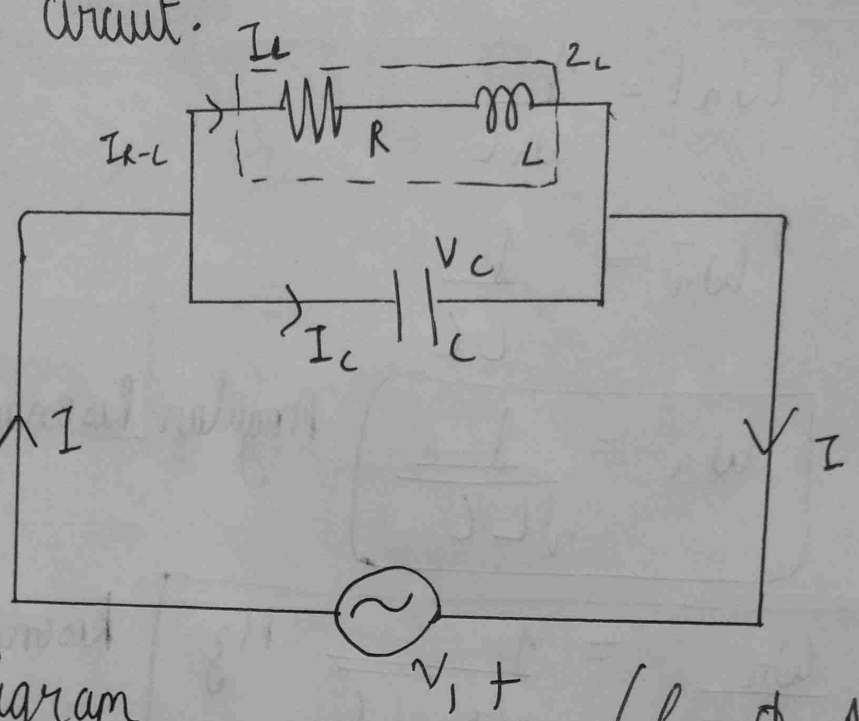
$$Q = \frac{V_L}{V} \text{ or } \frac{V_C}{V}.$$

$$Q = \frac{I_r X_L}{I_r R} \Rightarrow Q = \frac{X_L}{R}$$

$$Q = \frac{W_r L}{R} = \left(\frac{1}{\sqrt{LC}}\right)\left(\frac{1}{R}\right)$$

$$\boxed{Q = \frac{1}{R}\sqrt{\frac{L}{C}}} \geq 1$$

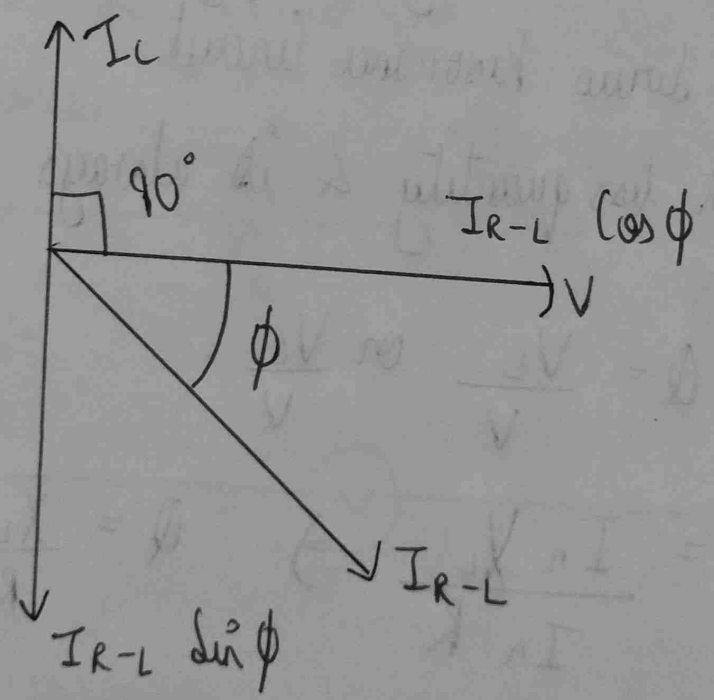
Q) Derive Resonance frequency & quality factor of parallel resonance circuit.



Circuit Diagram

$(\cos\phi = 1,\ \phi = 0^\circ)$

Phasor Diagram



For parallel Resonance to be take place I_{R-L}.

$$I_C = I_{R-L}\sin\phi$$

$$\frac{V}{X_C} = \left(\frac{V}{Z_{R-L}}\right)\left(\frac{X_C}{Z_{R-L}}\right)$$

$$Z^2_{R-L} = X_L X_C$$

$$Z^2_{R-L} = \frac{\omega_r L}{\omega_r C}$$

$$\boxed{Z_{R-L} = \sqrt{\frac{L}{C}}}$$

$$\sqrt{R^2 + X^2} = \sqrt{\frac{L}{C}}$$

$$R^2 + X_L^2 = \frac{L}{C}$$

$$X_L^2 = \frac{1}{C} - R^2$$

$$\omega_r^2 L^2 = \frac{L}{C} - R^2$$

Divide by L^2 on both the sides

$$\omega_r^2 = \frac{1}{LC} - \left(\frac{R}{L}\right)^2$$

$$\boxed{\omega_r = \sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2} \text{ Rad/sec}}$$

Angular frequency.

$$\boxed{f_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2}} \text{ Hz}$$

Resonance frequency.

Quality factor

It is the current magnification in parallel Resonance circuit

$$Q = \frac{I_L}{I} \text{ or } \frac{I_L}{I}$$

$$Q = \frac{V/Z_L}{VRC/L}$$

$$Q = \left(\frac{V}{Z_L}\right)\left(\frac{L}{VRC}\right)$$

$$Q = \left(\frac{V}{\sqrt{\frac{L}{C}}}\right)\left(\frac{L}{VRC}\right)$$

$$Q = \left(\sqrt{\frac{C}{L}}\right)\left(\frac{L}{RC}\right)$$

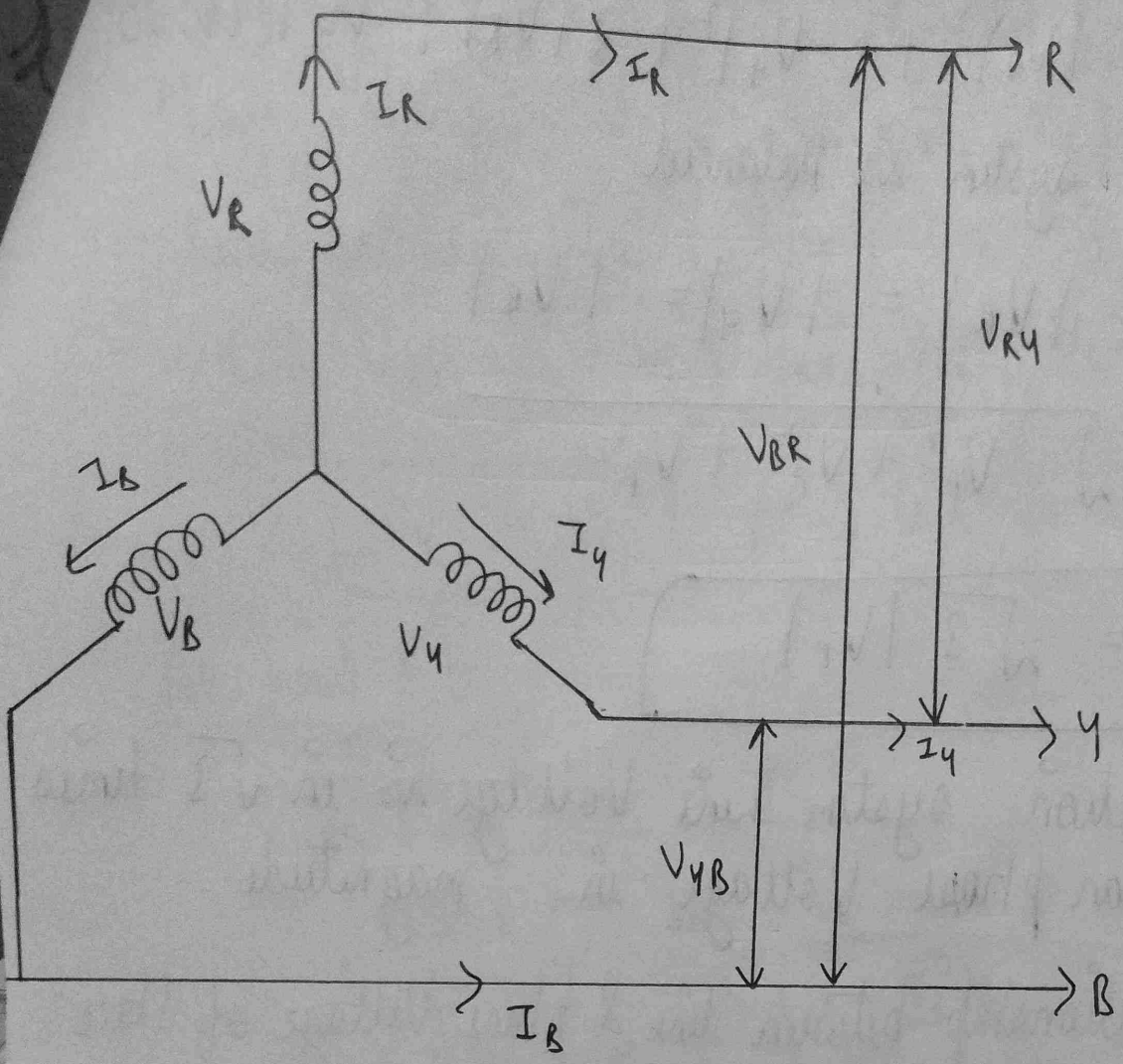
$$\boxed{Q = \frac{1}{R}\sqrt{\frac{L}{C}} \geq 1}$$

It is a unit less quantity. It is greater & less than 1.

Q) Derive Relationship between line & phase quantity of star connected system.

$$\frac{360}{3} = 120^\circ$$

Star Connection obtained by joining similar ends of the coil.



Circuit Diagram.

Phase Voltage V_P (V_R, V_Y, V_B)

Phase Current I_P (I_R, I_Y, I_B)

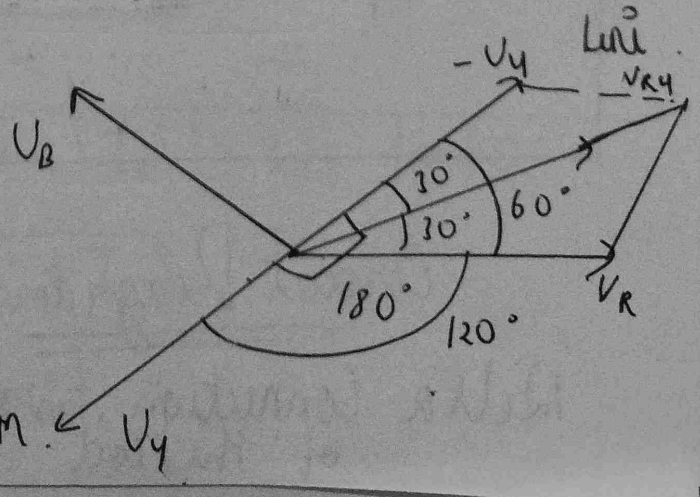
Line Voltage V_L (V_{RY}, V_{YB}, V_{BR})

Line Current I_L (I_R, I_Y, I_B)

$\boxed{I_L = I_P}$ ($\because$) Coil is connected in series with Line.

$\overrightarrow{V_{RY}} = \overrightarrow{V_R} - \overrightarrow{V_Y}$

$\overrightarrow{V_{RY}} = \overrightarrow{V_R} + (-\overrightarrow{V_Y})$



Phasor Diagram.

$$|\overrightarrow{V_{RY}}| = \sqrt{|V_R|^2 + |-V_Y|^2 + 2|V_R||-V_Y| \cos 60^\circ}$$

As system is balanced

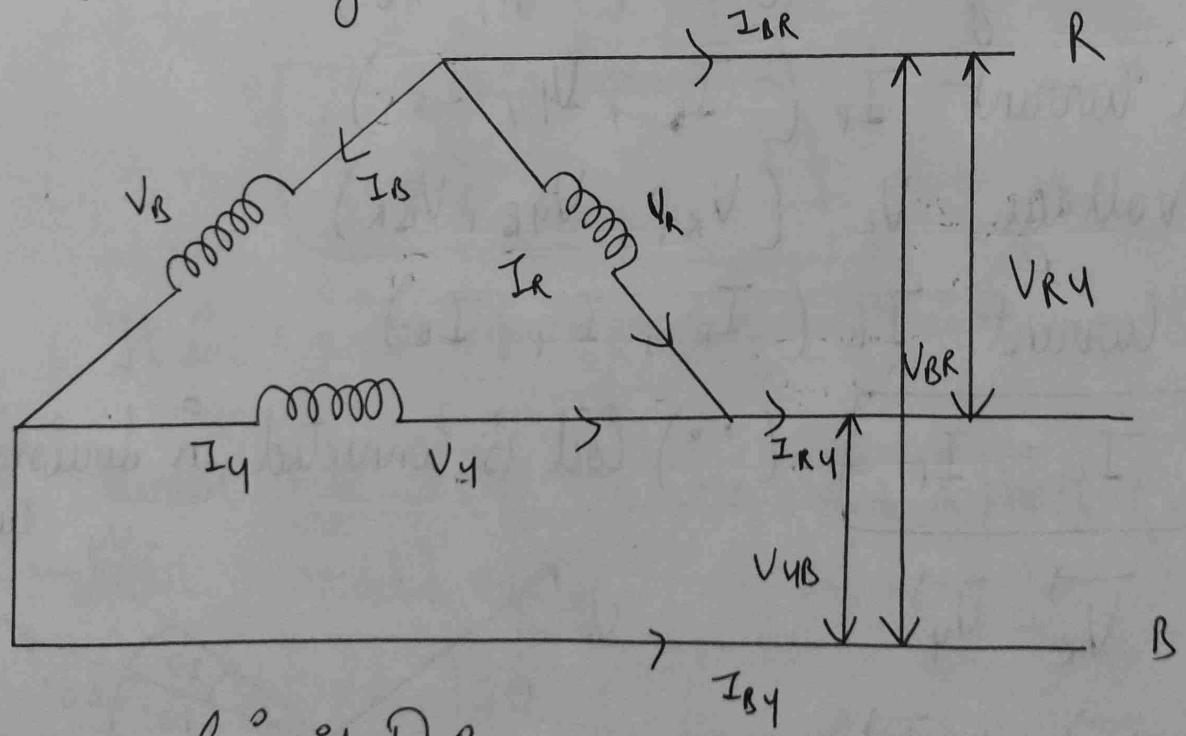
$$|V_P| = |V_R| = |V_Y| = |V_B|$$

$$|\overrightarrow{V_{RY}}| = \sqrt{V_P^2 + V_P^2 + V_P^2}$$

$$\boxed{|V_L| = \sqrt{3}\,|V_P|}$$

- A Star Connection system Line Voltage is in $\sqrt{3}$ times greater than phase Voltage in magnitude.

Q) Derive Relationship between line & phase Voltage of Star connected system.



Circuit Diagram

Delta Connection is obtained by joining dis similar ends of the coil.

Phase Current I_P (I_R, I_Y, I_B)

Phase Voltage V_P (V_R, V_Y, V_B)

Line Current I_L (I_{RY}, I_{BR}, I_{BY})

Line Voltage V_L (V_{RY}, V_{BR}, V_{BY})

$$\boxed{V_L = V_P}$$ (∵) Coil is connected in between two lines.

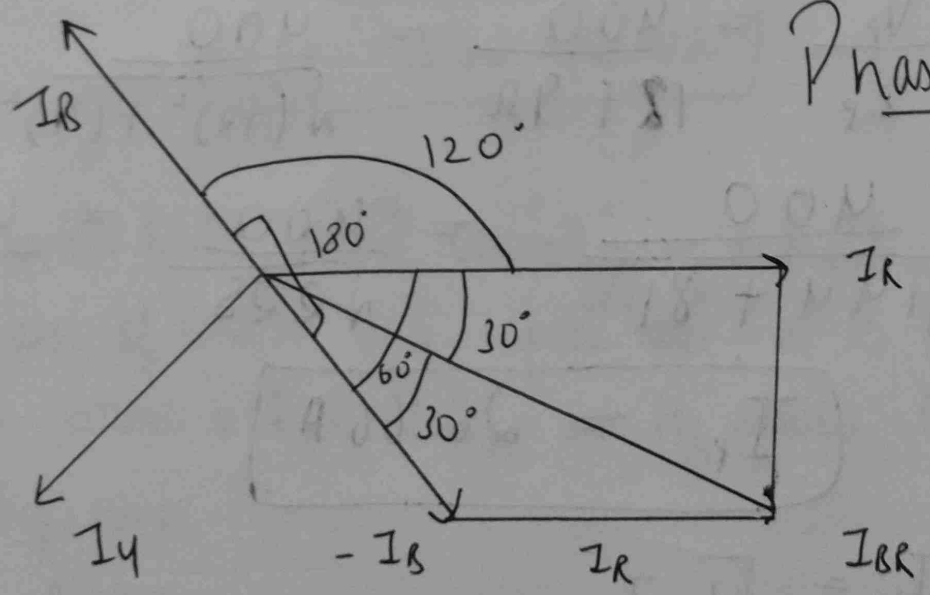
<u>At point B :</u>

$$I_R = I_B + I_{BR}$$

$$I_{BR} = \vec{I_R} - \vec{I_B}$$

$$\vec{I_{BR}} = \vec{I_R} + (-\vec{I_B})$$

<u>Phasor Diagram</u>



$$|I_{BR}| = \sqrt{|I_R|^2 + |-I_B|^2 + 2|I_R||-I_Y| \cos 60^\circ}$$

$$|I_{BR}| = \sqrt{I_R^2 + I_B^2 + I_R I_Y}$$

As system is balanced.

$$I_P = |I_R| = |I_Y| = |I_P|$$

In delta connected system the magnitude of line current is $\sqrt{3}$ times greater than the magnitude of phase current.

Q) A balanced delta connected load impedance $12+9\,\Omega$ per phase is connected to a three phase 400V supply. Find the line current, power factor, active power & total apparent power.

$$\boxed{V_L = 400V}$$

$$\boxed{I_L = \frac{V_L}{\sqrt{3}\, Z_P}}$$

$$Z_P = (12+9)\,\Omega \,/\, \text{Phase}$$

$$I_P = \frac{V_P}{Z_P} = \frac{400}{12+9\Omega} = \frac{400}{\sqrt{(12)^2+(9)^2}}$$

$$= \frac{400}{\sqrt{144+81}} = \frac{400}{\sqrt{225}} = \frac{400}{15}$$

$$\boxed{I_P = 26.66\,A}$$

$$I_L = \sqrt{3}\, I_P$$

$$I_L = \sqrt{3}(26.66)$$

$$I_L = 46.18802\ A.$$

$$\cos\phi = \frac{R_P}{Z_P} = \text{Power factor}$$

$R_P = 12\,\Omega$

$X_P = 9\,\Omega$

$$\cos\phi = \frac{R_p}{Z_p} = \frac{12}{15} = 0.8$$

Active Power

$$\boxed{P = \sqrt{3}\, V_L\, I_L \cos\phi}$$ formula.

$$P = \sqrt{3} \times 400 \times 46.2 \times 0.8$$

$$\boxed{P = 25600 \text{ Watt}}$$

Total Apparent Power.

$$S = VI$$

$$S = 400 \times 46.2$$

$$\boxed{S = 19200 \text{ VA}}$$

Q) A series RLC Circuit consisting of a Resistance of 20 Ω, inductance of 0.2 H and capacitance of 15 micro farad is connected across a 230 Volt, 50 Hz source Calculate :-

(I) Impedance

(II) Current

(III) frequency of supply to be adjusted to make power factor unity

(IV) Quality factor.